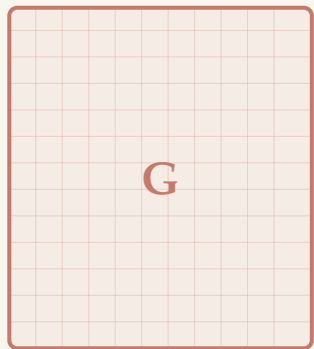
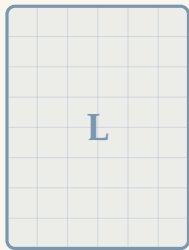


Kronecker Factorization of the Preconditioner



$mn \times mn$
intractable
 $O(m^2n^2)$ storage

$\approx$



$m \times m$
 $O(m^2)$ storage



$n \times n$
 $O(n^2)$ storage



Shampoo savings:

$O(m^2 + n^2)$
vs $O(m^2n^2)$